Abhisekh Kumar

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Profile Summary

Software Engineer working on building scalable backend systems, real-time applications, and RESTful APIs using Java, Python and TypeScript and designing, and deploying. Leveraging LLM-based AI systems using GPT, RAG architectures, and transformer models to integrate AI into systems while continuously learning in modern software engineering practices.

Skills

Programming & Systems: Python, Java, C/C++, JavaScript, TypeScript, SQL, Data Structures Algorithms, API Development, Data Handling and security

AI/ML Tools: Tensorflow, Keras, Scikit-learn, PyTorch, Deep Learning, NLP, LSTM, OpenCV,Numpy, Pandas, Model training,Evaluation and validation, Matplotlib and statstics.

Developer Tools: Git, GitHub, Anaconda, Postman

Experience

Botsquad || Software Engineer **Nov 2024 - Present**

Python || SQL || LLM || Java || API || RAG

- Integrated AI model for dynamically adjusting obstacle density, speed, and hazard frequency based on player behavior, resulting in a 21% increase in player retention.
- Designed player analytics pipelines processing 100+ gameplay events/day, enabling pattern detection on game play of player which contributed to a 17% improvement in average session length.
- Built context-aware procedural challenge generation logic, allowing real-time difficulty scaling based on player skill, reducing early-stage player drop-off by 24%.
- Integrated AI-assisted scoring and reward optimization logic, balancing coin drops and score multipliers using player engagement data, leading to a 19% increase in repeat play sessions.
- Collaborated with backend teams on RAG + LLM-assisted analytics, leveraging aggregated gameplay data to generate engineer-facing tuning insights and player-facing hints/dialogues.

Samsung Research and Development || Internship **Nov 2022 - Jun 2023**

Tensorflow || Keras || Numpy || One Hot Encoding || Deep Learning Model|| NLP

- Built **Phonetic Model for Indian Languages** translating **4 Indian languages into English phonemes**, achieving an **F1-score of 81.0** using **LSTM**.
- Improved performance by **2.5% over SOTA baselines**, reducing phoneme recognition errors by **15%**.
- Processed and pre-cleaned **4 lakh multilingual phoneme pairs**, ensuring dataset reliability.

Projects

Generative AI Based Video Generation for Classical Dance (R&D) **2023 - 2024**

Python || ML Model || BLIP2 Pre-processor

- Designed pipeline to generate **realistic Bharatnatyam dance sequences from text**, integrating BLIP2 for motion understanding.
- Processed **5K+ motion capture frames**, improving realism of avatar movements.

Gesture Driven Autonomous Flight of Drones (Final year project) **2024**

Mediapipe || Tensorflow || OpenCV || DJITelloPy

- Built a gesture-driven drone navigation system with **90% recognition accuracy**.
- Enabled commands (take-off, hover, forward/backward, landing) with **200ms latency**.
- Tested in **200+ real-world flight sessions**, ensuring robust navigation.

RAG Based Chatbot **2025**

Python || RAG || JavaScript || HTML/CSS

- Developed a RAG-based chatbot using a **1,000-row domain dataset** and Gemini LLM, achieving **~95%** contextual relevance through semantic retrieval and prompt grounding.
- Implemented a retrieval-augmented LLM pipeline that reduced hallucinated responses by **~28%** compared to prompt-only chatbots and Optimized retrieval and prompt logic to maintain **90%** response accuracy.

Object Detection App (Mobile Application) **2023**

Java || Tensorflow Lite || Android Studio

- Developed Android app for real-time object detection, achieving **92% accuracy**.
- Integrated Webview to provide on-device object insights with full **privacy preservation**.

Education

Cambridge Institute of Technology

B.E. in AI&ML || CGPA: 7.9/10 - 71.9%

Kendriya Vidyalaya ASC Centre (CBSE Curriculum)

Class 12 - 80.8% || Class 10 - 88.4%